

WILD CONDUCTOR EXPONENTS OF CURVES: ERRATA

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- Theorem 1.4/6.5 and Lemma 6.4 need the additional assumptions that the leading coefficient of f as a polynomial in x lies in $K^\times$ and that the genus is constant under the perturbations described.¹
- The proof of Corollary 1.7/6.6 should also note that the extra assumption on genera is satisfied because a curve $y^n = f(x)$, f square-free, has genus

$$g = \frac{(n-1)(\deg f - 1) - (\gcd(n, \deg f) - 1)}{2}.$$

- The $R_q = 0$ paragraph in the proof of Proposition 4.2 should read “all of $\ker \pi_*[1 - \gamma]$ ” instead of “all of $\ker \pi_*[q]$ ”.
- The specialisation argument in the sentence “We can ensure this is the case...” in the proof of Theorem 1.4/6.5 should be replaced by a reference to Lemma 5.2 here². At the end of the same proof, “ ℓ -adic” should read “ p -adic”.
- Lemma 5.7 should be replaced by the following:

Lemma 0.1. *Let $X \rightarrow B$ be an S_n -cover over a finite extension of $\mathbb{Q}_p$, with $p > n$, such that $\pi: X/S_{n-1}^\circ \rightarrow B$ is simply branched, except possibly above one geometric branch point. Let $C_m \leq S_n$ contain no transposition. If π is simply branched, then the cover $X \rightarrow X/C_m$ is unramified.*

If π is not simply branched, then wild inertia acts trivially on the geometric points of X above the exceptional fibre. In particular, for every prime $q \mid m$, wild inertia acts trivially on the ramification points of $X \rightarrow X/C_m$ whose ramification index is divisible by q .

Proof. Fix a geometric point $P \in B$, and choose a point $\mathfrak{P}$ on X lying above P . The points above P on X/S_{n-1}° correspond to the orbits of $\{1, \dots, n\}$ under the action of the inertia group at $\mathfrak{P}$. The assumption that $X/S_{n-1}^\circ \rightarrow B$ is simply branched away from a particular geometric branch point thereby implies that, as we vary over P , each inertia group is either trivial or generated by a transposition, except for those above this geometric branch point.

The inertia group of $X \rightarrow X/C_m$ at $\mathfrak{P}$ is the intersection of C_m with the inertia group of $X \rightarrow B$ at $\mathfrak{P}$. Because C_m contains no transposition, this cover is unramified away from any exceptional fibre.

In the non-simply branched case, write b for the exceptional geometric branch point and let $I \leq S_n$ be the inertia group at a chosen point $\mathfrak{P}$ of X above b . By uniqueness, b is fixed by G_K . We identify geometric points of X above b with S_n/I . The action of G_K commutes with the S_n -action, so the induced action of G_K on S_n/I factors through the group N_I/I , where N_I is the normaliser of I in S_n , of order prime to p . Therefore, W_K acts trivially on S_n/I , so on the points above b . The final claim follows because the ramification points of $X \rightarrow X/C_m$ above b form a subset of this fibre. $\square$

¹A stronger proof allows the leading coefficient $a_n(y)$ to simply need $w_K(a_n(y)) = 0$ and no restriction on the genera, cf. Corollary 1.3 of <https://arxiv.org/abs/2607.24454>.

²G. van der Geer and A. Kouvidakis. “The class of a Hurwitz divisor on the moduli of curves of even genus,” *Asian J. Math.*, 16(4):787–806, 2012